

Rate Card Upload — Final Build & Design Specification

Audience: Design (Claude Design) + engineering. This is the design-handoff version — it carries both the functional build and the design instructions for every screen. **Scope:** Everything from the moment the user **initiates an upload** to the new rate cards **going live**. **Flow:** ① Upload → ② Validate → ③ Rate Model (3.1 Pay Rate Engine · 3.2 Agency Rate Configuration) → ④ Agency Rate Cards → ⑤ Activate (Go Live). **Status:** Final draft. Supersedes the working build spec; the five component specs remain as detailed references.

How to use this spec (design). §1 is the shared design system — apply it on every screen. §2-§4 are the rules the UI must express. §5-§9 are the five steps; each gives **Purpose · Layout · Components & behaviour · States**. Wireframes are **structural intent, not final visuals** — design owns the visual language within the §1 conventions; the labelled elements and hierarchy are the requirements.

1. Shared design system & conventions (*apply to every screen*)

- **Stepper.** Persistent 5-step progress at top: (① Upload ✓ – ② Validate – ③ Rate Model – ④ Agency Rate Cards – ⑤ Activate). Completed steps ticked, current highlighted.
- **Severity language (consistent everywhere):** (🚫 Blocker) (red), (? Confirmation) (blue/neutral), (⚠ Warning) (amber), (✓ OK / success) (green), (i Info). Row status uses a coloured dot with an issue count.
- **Bill rate is the primary visual anchor.** Wherever a bill rate appears it is the largest/boldest figure in its row or card; supporting numbers (pay rate, burden, markup, margin %) are secondary.
- **Three cell treatments — always visually distinct:**
 1. **Editable input** — clear affordance (bordered/fillable).
 2. **Derived/computed** — plain, non-interactive (e.g. Basic+Geo, burden amounts, bill rate).
 3. **Locked statutory** — non-interactive with a (statutory · engine vX) tag; no input affordance.
- **Tier chips on every waterfall line:** (Pay) (reference) · (Engine) · (Agency), one consistent colour per tier, so any rate is auditable at a glance.
- **Expandable rows everywhere.** Any rate/line expands to its **tier-tagged waterfall** ((pay rate → burden lines → markup = bill rate)). One consistent expand affordance.
- **Change/delta styling.** Δ£ and Δ% are colour-coded by direction (increase vs decrease) with directional arrows; "what changed" shown as small attribution chips.

- **Version badges:** Active · Pending · Scheduled · Superseded. **Region-tier badges:** National · London.
 - **Gated primary buttons.** The forward/commit button is disabled until the step's exit conditions are met, with a tooltip stating what's outstanding (e.g. "2 blockers to resolve").
 - **Drill-down principle.** Summary rows stay narrow; detail lives in expansion or a side panel — never widen the table to fit everything.
 - **Live recompute feedback.** Editing any input shows the recomputed result (old → new) on affected outputs immediately.
 - **Standard states per surface:** loading/processing (with a labelled progress sequence where relevant), empty, error, read-only (no-permission).
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2. VMS terminology (*the rate vocabulary the UI must use*)

- **Pay rate** — what the worker is paid (the workbook's "net pay"; **£12.21 is the NMW floor**, not a constant — varies by tier/site/role/parity/day-type).
- **Statutory burden** — employer on-costs added to pay (WTR holiday, Employer NI, pension, levy, sick pay). Pay rate + burden = **fully-burdened pay rate**.
- **Markup** — the flat £/hr a supplier adds to the burdened pay rate.
- **Bill rate** — billed to the client = **pay rate + burden + markup**.
- **Margin %** — *derived metric*: $\text{markup} \div \text{bill rate}$ (a read-out, never an input).
- **Supplier** $\equiv$ **Agency** (UI says "Agency"); **Rate card**; **Labor category** $\equiv$ **job type / position group**; **Effective date**.

Core formula the UI expresses end-to-end: **Bill rate = Pay rate + Statutory burden + Markup.**

3. Governing build principles

1. **Staging, never live.** All work is on a staged copy; live bill rates untouched until Activate (⑤).
2. **Recompute from source — never trust the file's computed bill rates.** Recompute from **standardized** burden formulas + parameters; the workbook's per-row formulas are inconsistent (some pension rows drop the £967 cap; one is a bare $\text{pay} \times 1.5\%$) and are **normalized, not imported**. Surface where recomputed $\neq$ uploaded.
3. **One owner per number.** Pay rate $\rightarrow$ Pay Rate Engine; statutory burden **rates** $\rightarrow$ engine constants; supplier markup & parameters $\rightarrow$ Agency Rate Configuration.
4. **Nothing disappears silently.** Every uploaded row is classified or listed as **skipped, with a reason**.

5. **Everything is audited.** Mappings, confirmations, edits, effective date, cutover policy, activation.
 6. **Statutory compliance is enforced, not assumed** (§8.3 / §9.4).
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4. Target data model (what the upload populates)

- **Engine constants (statutory, shared):** WTR **12.07%** (**14.04% parity**); Employer NI **15%** above **£96/wk**; pension qualifying-earnings band **£120-£967/wk**; levy **0.5%**; peak factor.
 - **Pay Rate Engine:** pay rate keyed by **Region tier (National/London) × Site × Labor category × Parity × Day-type** + the burden formulas → the fully-burdened pay rate.
 - **Agency Rate Configuration:** per **supplier × position group — markup (£/hr), pension %, weekly hours, sick pay (£/hr or %), levy inclusion (Y/N)**.
 - **Bill-rate chain:**

pay rate → +WTR → +Employer NI → +pension → +levy → +sick =
fully-burdened pay rate → +markup = bill rate
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5. Step ① — Upload

Purpose. Receive the workbook and confirm it's parseable.

Layout. Drop-zone / file-picker; accepted-format note; upload progress; on submit, a **readability check** ("checking the file..."). On failure, an inline error stating what's wrong with a route back to re-select.

States. Idle → Uploading → Readability check → {Pass → Validate · Fail → reason shown, stay on Upload}.

6. Step ② — Validate (*is the data sound and organized?*)

Purpose. Show every warning, issue, and update; let the user make the **data-quality** fixes. (Supplier-parameter edits happen in ③.2.)

Layout.

```

└ Stepper ────────────────────────────────────────────────────────────────────┘
| ✓ We read 132 rates from Rate_Card.xlsm                                     |
| 🚫 2 to fix      ? 4 to confirm    ▲ 11 to review                          |
|                                                                              |
| A · How we organized your data                                             |
|   Pay-rate data · Agency-config data · constants · lookups              |
|   ▶ Column mapping                ▶ Skipped rows (each with a reason)    |
|                                                                              |
| B · Resolve before continuing                                              |
|   🚫 Blockers (2)                  ? Confirmations (4)                    |
|                                                                              |
| C · What this upload will change                                           |
|   +6 new · ~18 changed · -2 removed    ▶ per-row diff ( $\Delta f$  ·  $\Delta \%$  · why) |
| └ Back to upload ─────────────────── Continue to Rate Model (gated) ─────┘

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Processing view first shows a labelled sequence: **Parse** → **Map** → **Classify** → **Normalize** → **Validate** → **Diff**.

Components & behaviour.

- **A — Organization (transparency).** Classification counts into Pay-Rate-Engine data / Agency-Config data / engine constants / lookups; **expandable** (source column → model field) mapping (editable, with auto-mapped/confidence indicator); **expandable** skipped-rows list, each with its reason.
- **B — Resolve. Blockers** (must fix) each show what's wrong, which rows, and an **inline resolution control**. **Confirmations** are small explicit choices with a **default pre-selected** and the impact shown.
- **C — Updates.** Dataset-level change summary; **expand to a per-row diff** (old → new, $\Delta\text{£}/\Delta\%$ colour-coded with arrows, and "what changed" chips — (Pay rate ↑), (Markup ↑), (Hours), (New mapping)).

States. Processing · Parse-failure (→ Upload) · Blockers/Confirmations present (Continue gated) · Warnings-only (Continue enabled) · Clean.

7. Step ③ — Rate Model (*the calculation, in two parts*)

7.1 Pay Rate Engine (read-only — how pay rates are calculated)

Purpose. Let the user verify the shared, statutory foundation before any supplier pricing.

Layout.

[Region tier: 'National' London]

Site: [Warrington ▼]

Columns: 'Net pay by day-type' Rate build Premiums & shifts

Role (grouped by job type)	Parity	Std	Fri	Sat	Sun
Warehouse					
Warehouse Operative AM	Pre	12.21	12.21	12.49	12.49
Warehouse Operative PM	Pre	12.32	12.32	12.75	12.75
Warehouse Parity					
Warehouse Operative AM	Post	12.28	12.28	12.57	12.57

► expand a role → cost-side waterfall: pay → +WTR → +NI → +pension → ... = fully-burdened pay rate

Components & behaviour.

- **Scoped by location** — Region-tier toggle (National | London) + Site selector above the grid; optional side-by-side compare (*National vs London, vs previous version*).
- **Rows = roles grouped by labor category; parity is a row attribute (Pre/Post visibly paired)** so the uplift reads at a glance.
- **Switchable column groups** to control width: *Net pay by day-type* (Std/Fri/Sat/Sun/OT) · *Rate build* (Hourly rate/Geo/Basic+Geo) · *Premiums & shifts*.
- **Read-only** (corrections to pay go via re-upload; statutory rates via engine admin) — render all cells in the derived/locked treatments (§1), not as inputs.
- **Cost-side waterfall** on expand (pay → burden = fully-burdened pay rate), tier-tagged.
- **Verification cues:** NMW-floor markers (distinguish a genuinely-floored £12.21 from one that should be higher), a flag where **London ≤ National**, and **gap highlighting** (missing day-type / missing role at a site).

States. Loading · grid · compare · empty (no pay rates for the tier/site).

7.2 Agency Rate Configuration (*editable — supplier overrides & markup*)

Purpose. Configure each supplier's markup and cost parameters.

Layout — two view modes. *By supplier (primary — configure one supplier)*: a table keyed by position group:

Supplier: [Staffline Group ▼]

Position group	Markup £/hr	Pension %	Hours /wk	Sick pay £/hr or %	Levy incl.
Warehouse	0.40	1.5	37.5	— (none)	Y
Transport	1.05	1.9	40	£0.06	Y

Statutory burden (read-only): WTR 12.07% (14.04% parity) · Employer NI 15% >£96/wk
· levy 0.5% · pension band £120–£967

Compare suppliers (secondary — peer view): one matrix per parameter, **rows = supplier**, **columns = labor category**, grouped **Commercial** (Markup) vs **Cost inputs** (Pension % · Hours · Sick · Levy), so blanks (missing markup) and outliers jump out.

Components & behaviour.

- **Editable cells** (markup £, pension %, hours, sick pay = value + a **fixed-£/%/none type selector**, levy Y/N toggle) in the editable treatment; **statutory burden shown read-only alongside** (§1 locked treatment).
- **Margin vs cost-inputs framing:** markup is a commercial markup; the rest are cost inputs that re-run the burden calc — reflect this grouping visually.
- **Live recompute** on edit (preview the affected supplier's bill rates).
- **Bulk & reuse:** "apply to all position groups", apply down a column across suppliers, **clone a supplier's whole config**.
- **Validation (inline):** missing markup → invalid; margin/pension/hours outside peer band → outlier warning; blank-vs-zero confirm on sick/levy.
- **Not editable here:** pay rate (③.1), statutory **rates**, derived burden amounts. To "fix" a pension figure, edit the **pension %**.

States. Per-supplier edit · compare matrix · recompute-in-progress · read-only (no-permission).

8. Step ④ — Agency Rate Cards (*the output + simulation*)

8.1 Rate cards by supplier

Purpose. Show the resulting bill-rate cards.

Layout.

[Group by: 'Supplier' Site Labor category] [Filter ▼] [🔍] chips: Errors · Missing markup · Outliers · Changed

● Staffline Group ▶							
Market	Labor category	Pay	Burden	Markup	BILL RATE	Margin %	
Warrington	Warehouse	12.21	3.65	2.25	■ 18.11 ■	12.4%	▶
↳ expand → waterfall: pay → +WTR → +NI → +pension → +levy → +sick → +markup = bill rate (tier-tagged)							

Bill rate is the row anchor (boldest). Status dot + issue count per row; flagged waterfall lines show their note inline.

8.2 Simulator — "what would we actually bill?"

Purpose. Resolve a concrete scenario from the **staged** cards to confirm real billing.

Layout — two columns.

LEFT (inputs)	RIGHT (outputs)
└ Pricing config ─────────┐	└ Hourly bill rate ─────────┐
Position · Supplier ·	Pay rate £12.21 /hr
Legal entity · Location	+ WTR (12.07%) £1.47
└ Worker ─────────┐	+ Employer NI £1.69
Age · Tenure (→parity) ·	+ Pension £0.41
Hours worked this week	+ Levy · + Sick
└ Assumptions ─────────┐	= Fully-burdened pay £15.86
locked statutory (read-only)	+ Markup £2.25
editable supplier params	= BILL RATE ■ £18.11 ■
VAT · OT threshold · OT rate	└ This booking ─────────┐
	bill × hours + VAT = total incl VAT
Presets: highest-markup · parity ·	tiles: £/hr incl VAT · margin % ·
under-21 · each new/changed line	bill-vs-pay multiple

Components & behaviour.

- **Inputs grouped:** Pricing config (Position · Supplier *selectable* · Legal entity · Location → resolves which pay rate); **Worker** (Age · Tenure → derived parity · hours-worked); **Assumptions** (locked statutory in the locked treatment; editable supplier params; commercial VAT/OT); **Booking** (dates/times → hrs/shift · days · total).
- **Outputs:** the tier-tagged hourly **bill-rate waterfall**; **booking total** (bill × hours + OT + VAT = total incl VAT); **summary tiles** (bill/hr incl VAT · margin % · bill-vs-pay multiple).

- **Presets** for the riskiest cases; a scenario can be **pinned**.
- **UK-law guardrails (§8.3)** surface inline as warnings/errors during simulation.

8.3 UK-law guardrails (full table in *Rate_Card_Simulator_Spec.md* §7)

Age drives three things the static workbook misses; **Tenure** drives parity:

- **NMW by age** (NLW 21+ £12.21; 18-20 £10.00; 16-17/apprentice £7.55) — warn if pay rate < the age-band floor.
- **Employer-NIC relief** for under-21s/apprentices under 25 (0% up to £50,270) — their Employer NI ≈ £0; flat 15% overstates.
- **Pension auto-enrolment** (22→SPA, >£10k; employer min 3% of qualifying earnings) — **the file's 1.5-2.5% supplier rates are below this minimum → flag likely non-compliant**; suppress pension for non-eligible workers.
- **AWR parity at 12 weeks** → post-parity pay rate **and** parity WTR (14.04%).
- **Levy** only if supplier paybill > £3m; **SSP** is a real cost (0% understates); **VAT 20%** on the whole bill; **Working time** 48-hr average — warn if exceeded.

Statutory figures are 2025/26; source centrally and re-verify against current HMRC/DWP tables each April. Monitor (don't hard-code) Employment Rights Bill day-1 SSP and umbrella-company regulation.

8.4 Exit. Review + simulate, then **Continue to Activate**. No live data written here.

9. Step ⑤ — Activate (Go Live) (*the only step that writes live bill rates*)

Purpose. Publish the staged cards with an effective date and manage the cutover safely.

Layout.


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└ Effective date _____
| ( ) Immediate      (•) Scheduled [ dd/mm/yyyy ] ▲ mid-pay-week warning|
└ Cutover for active work _____
| (•) New bookings only ( ) Apply forward to all active ( ) Per-contract|
|   → 24 active assignments affected by this choice                      |
└ Impact _____
| Active assignments: 24 · Bill movement: +£1,920/wk · ↑18 ↓4          |
| 🚫 1 rate decrease breaches NMW/parity at the effective date         |
└ Re-validation _____
| ✓ compliance re-checked on effective date · ▲ live version changed since|
|   Validate (re-diff) · engine v0.82 confirmed                        |
└ Approval _____ (if required) Pending → Approved _____
└ Back _____ Activate (gated) _____

```

Components & behaviour.

- **Effective date:** Immediate or Scheduled; **warn on a mid-pay-week date** (splits the weekly NI/pension thresholds). A future date supersedes the prior version from that date; prior version end-dated at effective – 1.
- **Cutover policy (the key change-handling control):** radio — *New bookings only* (default-safe, active bookings grandfathered) · *Apply forward to all active* · *Per-contract* — with the **count of affected active assignments** shown for the choice.
- **Impact assessment:** affected active assignments/POs; total bill-rate movement (£), # up/# down, largest movers; every **rate decrease** flagged — any breaching **NMW or AWR parity** at the effective date is a **blocker**.
- **Re-validation at commit:** re-run compliance against the figures **effective on the chosen date** (NMW/NIC/SSP/pension band change each April); **re-diff against the current live version** (warn if it moved since Validate); confirm engine version.
- **Authorization:** permission-gated; optional **Pending approval → Approved → Activated** with the approver shown the impact.
- **Apply (commit): atomic** — all-or-nothing, roll back on failure (staged version preserved), concurrency-safe.

Post-activation. Confirmation summary (what's now Active Scheduled), effective range, headline impact); notifications to affected stakeholders; full audit record (effective range, engine version, cutover policy, approver, who); **rollback** to the previous version (window/permission); a Scheduled version is editable/cancellable before it goes live.

States. Ready · Blocked (NMW/parity breach, unresolved drift) · Pending approval · Applying (atomic) · Activated · Scheduled · Apply-failed (rolled back).

10. Consolidated validation taxonomy

Step	Type	Checks	Effect
② Validate	Blocker	Unreadable file / missing column / row missing required field / unmatched location or labor category / duplicate key / engine-constant mismatch	Must resolve
② Validate	Confirmation	Low-confidence mapping / blank-vs-zero / inconsistent levy / engine-constant acceptance / new site or labor category	Must answer (defaulted)
② Validate	Warning	Pension deviated from standard (recomputed) / rounding drift / hours mismatch / outlier / missing day-type	Review, non-blocking
③.2 Agency Rate Config	Value	Missing markup / pay rate missing or zero / bill rate $\leq$ pay rate	Block or acknowledge
④ Rate Cards (simulator)	Compliance	Below NMW (error) / pension < 3% (warn) / NIC relief not applied (warn) / parity not reflected (warn) / working time > 48h (warn) / levy threshold mismatch (info)	Surface at review
⑤ Activate (commit)	Cutover / drift	Rate decrease breaching NMW/parity at the effective date (blocker) / live changed since Validate (warn) / engine constants changed since staging (re-acknowledge) / statutory figures differ on a future date (apply the date's figures) / mid-period effective date (warn)	Final gate before live

Structural → **Validate**; value → **Agency Rate Configuration**; compliance at review → **simulator**; cutover/drift/final compliance → **Activate**.

11. State machine

State	Transitions
File received → Readability check	pass → Validate · fail → Upload (reason)

State	Transitions
Validate: Processing → {Blockers/Confirmations Warnings-only Clean}	resolve → Rate Model
Rate Model: Pay Rate Engine (review) → Agency Rate Configuration (edit)	value gate → Rate Cards
Agency Rate Cards: review → simulate	→ Activate
Activate: date + cutover → Impact & re-validation → {Ready Blocked Pending approval}	resolve/approve → Apply
Apply: Applying (atomic) → {Activated Scheduled Apply-failed (rolled back)}	live / scheduled / retry
Any step → Back / Cancel	discards staged data (confirm)

12. Audit & versioning

- Each step writes to the audit log: Validate confirmations + fixes; Agency Rate Configuration edits (field · old → new · user · time); effective date; cutover policy; approval; activation.
- **Activate creates a new version** (effective range, engine version, cutover policy, applied-by, change summary); prior version superseded from the effective date.
- Staged data discarded on cancel; Scheduled versions visible and editable/cancellable; rollback reverts to the previous version.

13. Acceptance criteria (whole flow)

- ☐ Shared design conventions (§1) are applied on every screen: stepper, severity language, bill-rate anchor, three cell treatments, tier chips, expandable waterfalls, Δ colour-coding, version/tier badges, gated buttons.
- ☐ Upload performs a readability check and routes failures back with a reason.
- ☐ Validate parses and either classifies or skips (with reason) **every** row; mapping shown and editable; blockers/confirmations gate Continue; warnings pass through; updates shown with a per-row diff.
- ☐ Bill rates are **recomputed from standardized formulas**, not imported; deviations flagged.
- ☐ Pay Rate Engine: read-only grid by tier/site/category, parity paired, switchable column groups, NMW-floor markers, cost-side waterfall.

- ☐ Agency Rate Configuration: per-position-group editable table (+ compare-supplier matrix); statutory burden read-only; live recompute; bulk/clone; value validation.
 - ☐ Agency Rate Cards: bill-rate cards by supplier with tier-tagged waterfall; two-column simulator with the §8.3 guardrails firing; selectable supplier; staged data only.
 - ☐ **Activate is the only step that writes live data:** requires effective date + cutover policy + impact assessment + passing re-validation; rate cuts breaching NMW/parity blocked; atomic commit with rollback; result is Active or Scheduled, audited.
 - ☐ Statutory figures sourced centrally; a future-dated activation uses the figures effective on its date.
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14. Open questions

- **Cutover default** — "new bookings only", or follow supplier contract terms by default?
- **Mid-period changes** — hard-block off-boundary effective dates, or warn and allow?
- **Rollback window** — anytime, or fixed window post-activation?
- **Approval workflow** — always required to publish, or only above an impact threshold?
- **Hard-block vs acknowledge** on value errors (③.2).
- **Editable scope** — confirm pay rate is import-only (corrected via re-upload), not editable in the Pay Rate Engine.
- **Comparison baseline** — always current live version, or any prior version?
- **Compliance source of truth** — hold NMW bands, SSP, NIC thresholds, QE band centrally with the effective tax-year surfaced.
- **Peak/bank-holiday cards** — apply the peak factor as a final waterfall line.